Henry Wang

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EDUCATION

Stanford University **Stanford, CA**

B.S. Math | Expected Graduation: June 2029 GPA: 3.92

- **Relevant Courses:** Machine Learning (CS 229), Efficient Generative Language Models, Decision Making under Uncertainty, Probability for Computer Scientists, Game Theory, Data Structures and Algorithms

EXPERIENCE

Stanford Safe and Intelligent Autonomy Lab | Drone Avoidance Demo **June 2026 - October 2026**

Student Research Assistant

- Built a camera-only safety filter to stop a drone navigation policy from crashing in a photorealistic 3D Gaussian-splat environment, improving flight completion from 5/30 (16.7%) to 25/30 (83.3%) randomized trials
- Fine-tuned DINOv2 as safety Q-function to predict Hamilton-Jacobi safety values of drone actions from camera images, trained over 10 DAgger rounds (100M+ labeled state-actions), improving R^2 from initial 0.44 to 0.945
- Raised the safety-score threshold to trigger earlier safety intervention, increasing flight completion from 66.7% to final 83.3% without retraining

Stanford Radio Club | Pepperwood Air Quality Dashboard **September 2025 - June 2026**

Data Analyst

- Analyzed air-quality data from radio-connected sensors to identify pollution patterns during prescribed burns; shared findings with local firefighters, co-authored research presented at AGU25, accepted at AGU26
- Used XGBoost and correlation analysis to limit recording to 20 seconds per minute, cutting sensor power consumption by 43% while maintaining correlation with continuous recordings ($r > 0.7$)

UNC Wearable Bioelectronics Lab | Blood Pressure Device Demo **June 2023 - April 2025**

Student Research Assistant

- Led all model selection (CNN-LSTM), data preprocessing, and training for predicting blood pressure from PPG and biometrics: systolic mean error +1.1, SD 4.8 mmHg, within medical-grade accuracy thresholds (AAMI/ISO)
- Validated device PPG against FDA-cleared clinical Vicorder PPG ($R^2 = 0.78$) and model robustness to Gaussian input noise (ICC > 0.7); saliency analysis showed significant contributions of height, weight, and age ($p < 0.001$)

PUBLICATIONS

- A. Gupta, **H. Wang**, C. Crosby, "Learned Visual Safety Filter for Collision-Free Drone Flight," *Learning for Dynamics & Control Conference (L4DC),* 2027 (submitted)
- T. Nguyen, **H. Wang**, et al., "A Pilot Human Study on Dual Photoplethysmography for Continuously Monitoring Pulse Wave Velocity and Blood Pressure," *IEEE Journal on Flexible Electronics*, 2025

PROJECTS

Tree-of-Thoughts for Math Theorem Proving via Lean | Python, Gemini API, Lean 4 | GitHub

- Built LLM math theorem prover with beam search over proof steps: Gemini 2.5 Flash proposes candidate tactics, an LLM scorer ranks them, and Lean 4 checks each step before the search expands it
- Solved 47 of 126 LeanDojo theorems (37.3%), versus 28 (22.2%) using the same model with up to five full-proof attempts (i.e. prompting: "formally prove this theorem") and verifier feedback, without model fine-tuning

TECHNICAL SKILLS

- **Languages:** Python, Java, C, MATLAB, SQL, R
- **Machine Learning Frameworks & Tools:** TensorFlow, PyTorch, scikit-learn, CUDA

LEADERSHIP & OTHER INTERESTS

Stanford Boxing Club | *Financial Officer* | 2026 Stanford Fight Night (my fight)

- Managed club finances and helped organize the inaugural 2026 Stanford Fight Night, which raised ~$5,000; currently planning the 2027 event and expanding participation to 5 additional colleges